

Esports and High Performance HCI

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ABSTRACT

Competitive esports is a growing worldwide phenomenon now rivaling traditional sports, with over 450 million views and 1 billion US dollars in revenue each year. For comparison, Major League Baseball has 500 million views and \$10 billion in revenue, FIFA Soccer 900 million and \$1.6 billion. Despite this significant popularity, much of the world remains unaware of esports – and in particular, research on and for esports is still extremely scarce compared to esports’ impact and potential.

The Esports and High Performance HCI (EHPHCI) workshop will begin addressing that research gap. In esports, athletes compete through the computer interface. Because this interface can make the difference between winning and losing, esports athletes are among the most expert computer interface users in the world, as other athletes are experts in using balls and shoes in traditional sports. The premise of this workshop is that people will apply esports technology broadly, improving performance in a wide range of human activity. The workshop will gather experts in engineering, human factors, psychology, design and the social and health sciences to discuss this deeply multidisciplinary enterprise.

CCS CONCEPTS

• **Hardware** → **Emerging interfaces**; • **Human-centered computing** → **Interaction devices**; **Pointing**.

KEYWORDS

esports, expert users, expert interaction techniques

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1 BACKGROUND AND RATIONALE

The growth of esports (competitive video gaming) in global industry and culture has been explosive over the last few years [3]. The esports audience generates over 450 million views and annual revenue of \$1 billion US [20]. This compares well to traditional sports such as Major League Baseball, which has 500 million views and \$10.7 billion [1]; and FIFA Soccer, which has 900 million and \$1.6 billion [22]. In 2019, viewership of the world championship of *League of Legends*, a popular esports title, was greater than that of the Super Bowl in the same year [21].

Like traditional athletes, competition drives esports athletes to excel. Players spend tens of thousands of hours in their chosen game, and often develop additional training regimens to master particular low-level tasks. Aim trainers [17, 26] are an example of special purpose tools built for training. Furthermore, entire suites have been created to help players optimize their higher level strategy [18].

As the limits of human performance are approached, ergonomics, precision, and latency of the player’s equipment play an increasingly larger role. For example, to improve latency (response time), athletes turn off computer graphics features like shadows, reflections, and antialiasing (edge smoothing). Even the slightest change in gaming peripheral devices, such as a choice of mousepad [2, 24], mouse weight [15] and shape [7, 19], position of displacement sensor on a mouse [11], mouse–display gain function [13], keyboard switch design [10, 16] are considered for achieving a better gaming performance and lowering fatigue from lengthy gameplay. The gaming industry strives continuously to provide the highest performance peripheral interfaces. Esports requires excellence not only in individual performance, but also team performance. Rich, remote team communication and collaboration are core elements of its events. Heads-up team displays and communication systems like Discord and Twitch have already gained attention from the broader tech community.

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2 CHALLENGES AND GOALS

Despite these significant trends and their relevance to the human-computer interface, related research is sparse [23]. Esports work to date has focused largely on team-building [5, 6] and the spectator experience [4, 12]. The EHPHCI workshop will begin addressing this lack with a focus on the potential of esports methods and technology to invigorate support for expertise in the human-computer interface, improving human performance more broadly. Toward this end, the workshop seeks to address at least the following challenges:

2.1 Low Latency Systems

As in most sports, quick response times are a fundamental advantage in esports. Maximizing this advantage requires low-latency interactive software and hardware [8, 25], with tightly integrated input, rendering and display components [9, 14]. Realizing such systems may drive fundamental changes in real-time interaction technology, reflecting deeper understanding of human perception and performance.

2.2 Expertise and Training

Effective training and team building/composition methodologies are still emerging in esports. Innovations in esports science help athletes train more effectively, with contributions from sports science, medicine and psychology. Technical opportunities include deeply instrumented assessments, analytics and biometrics, AI training partners, and specialized training interfaces.

2.3 Communication and Collaboration

As we noted above, remote team communication and collaboration are core to esports. As the pandemic continues and it changes the ways in which we work and educate, esports' tested technologies, techniques and experience will prove invaluable.

2.4 Social Impacts

Esports technology supports not only its athletes but also its organizers and its fans. Individually, esports faces challenges with its athletes, whose careers are often shortened by stress, injury, unhealthy team environments, and rapid change in the esports games themselves. The public is also concerned about the effects of prolonged use of esports games and technology more broadly. Collectively, esports supports large online communities, which face many of the same challenges as society as a whole: diversity, addiction, cheating and civility.

3 BENEFITS AND APPLICATIONS

We anticipate that research addressing challenges such as these will bring a range of broader benefits to society, much as research meeting the needs of film found wide application outside of that industry. Realizing these benefits will be important if esports research is to find public support and funding sources. One reason for our optimism is the pro-to-amateur product path of traditional sports: innovative equipment developed for professionals spreads to consumers, driving its demand and lowering its price. If esports

technology can do the same, then its new input, rendering, simulation and display, performance measurement, player and coach feedback, and training technologies will find many broader applications. We particularly seek the discussion of the following applications:

3.1 Productivity

Like esports athletes, knowledge workers spend thousands of hours using their tools. Unlike esports athletes, most knowledge workers expertise quickly reaches a plateau. Emerging esports technologies and lessons supporting experts might enable knowledge workers to improve their productivity further. Such technologies might include task-salient visuals and overlays, the surprising importance of low latency feedback and new vertically integrated systems to deliver it, and technologies for remote collaboration and teamwork.

3.2 Remote Operation

The current pandemic is only accelerating the development of remote technologies such as drones and telemedicine. As the military and health industries have already discovered, expertise in gaming is a good predictor of success in remote operation. With its drive to lower latency and compete in remote settings, esports technology is a natural fit for remote applications. Its emerging expertise in analytics and training should also prove particularly valuable.

3.3 Rehabilitation and Adaptive Interfaces

An adaptive esports community made up of players with physical and cognitive differences is rapidly emerging, driving rapid enhancements in the availability and complexity of adaptive esports equipment. This not only levels the playing field for adaptive athletes, but also opens novel frontiers for telerehabilitation and telemedicine.

3.4 Research on Human Performance

Because its field of play is completely synthesized, professional esports test the limits of human performance constantly, creating new questions about, understanding of, and quantitative measurement equipment for performance and perceptual science.

4 ORGANIZERS

The EHPHCI workshop's organizers span a range of disciplines (computer science, engineering, medicine, social science, analytics, and art), professions (academia and industry), and regions (Asia, North America, Europe, and Oceania). This will enable them to solicit and review the wide range of contributions they seek, particularly from Asia, where CHI will be located. The workshop's main contact is Benjamin Watson.

Benjamin Watson is Associate Professor of Computer Science at North Carolina State University. His Visual Experience Lab focuses on the engineering of visual meaning, and works in the fields of graphics, visualization, interaction and user experience. His work has been applied in entertainment, security, finance, education, and medicine. Watson co-chaired the Graphics Interface 2001, IEEE Virtual Reality 2004 and ACM Interactive 3D Graphics and Games (I3D) 2006 conferences, and was co-program chair of I3D 2007. More

recently, Watson co-founded the IEEE Visualization for Communication workshop series, now in its third year. Watson is an ACM and senior IEEE member. He earned his doctorate at Georgia Tech's Graphics, Visualization and Usability Center.

Josef Spjut is a research scientist of computing at NVIDIA. He completed a Ph.D. at the University of Utah and was a visiting professor of engineering at Harvey Mudd College in Claremont, California. His research interests include esports, ray tracing and GPU architecture. Josef's current research focus is understanding how game rendering impacts computer system characteristics like frame rate and latency. Combined with the understanding of how player performance is influenced by these system characteristics, his research seeks to advance the field of esports by allowing professional gamers to win thru skill in a balanced gaming environment.

Joohwan Kim is a vision scientist in Nvidia's New Experience research group based in Santa Clara, California. Joohwan received his Ph.D. from Seoul National University in 2009, and was a post-doctoral fellow in University of California, Berkeley until 2015. Joohwan's current interests are in understanding and improving viewer experience of various types of displays, especially regarding esports.

Sunjun Kim is an assistant professor in DGIST, South Korea, who is leading Smart Input Device Lab. He worked as a postdoc researcher in User Interfaces Group at Aalto University, Finland, from 2017 to 2020. He obtained his Ph.D. in Computer Science from KAIST in Feb 2017 under the supervision of Professor Geehyuk Lee. He published a number of papers in CHI and UIST conferences regarding performance optimization of input device design, text entry system and study, touch screen interaction, and haptic.

Jennifer Listman is Director of Research at Statespace Labs. She completed her Ph.D. in Anthropological Genetics at New York University and was an Associate Research Scientist in Psychiatric Genetics at Yale School of Medicine. Her research has combined large scale human biological data with behavioral data spanning genomics, human evolution, substance abuse, and education. Her research interests include biases in data collection and study design. At Statespace Labs, Jenny leads multiple internal and external research projects on gaming and esports behavior, performance, and training.

Raphael Wimmer is a lecturer at the University of Regensburg, Germany, where he leads the research group *Physical Affordances for Digital Media and Workflows*. He received his PhD in Human-Computer Interaction from the University of Munich (LMU). His research revolves around technical, ergonomic, and organizational aspects of human-computer interaction. Recent research focused on optical and capacitive sensing techniques and latency of input/output devices. He currently coordinates a research project on projected augmented reality in everyday contexts.

David Putrino is a physical therapist with a Ph.D. in Neuroscience from The University of Western Australia. David began his career as a clinician in Australia and then moved to the United States to study computational neuroscience at Harvard Medical School, MIT, and NYU. He has served as a faculty member at Weill-Cornell Medicine and Burke Medical Research Institute. David works to develop technology solutions for individuals in need of better healthcare accessibility. He works with many collaborators

including the High-Performance Division at Red Bull, Not Impossible Labs, and Leaders in Sports Performance. David is the author of "Hacking Health: How to make money and save lives in the HealthTech world", and his research has been featured on the ABC, Sport's Illustrated, the Wall Street Journal, the BBC, Time Magazine, Wired and the LA Times to name just a few. In 2019, he was named "Global Australian of the Year" in recognition of his contributions to healthcare.

Byungjoo Lee is an associate professor at KAIST. He has focused on modeling, simulating and optimizing human-computer interaction processes. Recently, he has proposed cognitive-mechanical models that predict user input performance in tasks given temporal constraints. The models can be applied to evaluate, optimize and improve the performance of esports players.

5 WEBSITE

A preliminary version of the EHPHCI website is up at ehphci.org, hosted at github.com. At this point, it is primarily an expanded call for participation. Once publications have been accepted, the website will link to accepted publications hosted on the Open Science Framework (OSF), enabling broad dissemination of workshop results.

6 PRE-WORKSHOP PLANS

Recruiting participants from a wide variety of communities will be key to the EHPHCI workshop's success. We expect good participation from academia, and have already begun building an email list of such participants. To reach a broader audience, EHPHCI organizers will contact pro- and exceptional amateur gamers on social network services (Twitter, Facebook) and within gamer communities, encouraging them to share their high-performance gaming experiences in position papers or case studies.

7 VIRTUAL WORKSHOP STRUCTURE

Most of the single-day (five hour) virtual EHPHCI workshop will be dedicated to presentations by authors of accepted papers, posters and case studies. We anticipate dedicating roughly three hours to presentations by approximately 12 authors, 30 minutes to a keynote speaker from the esports industry, and using the remaining time to allow participants to introduce themselves, formally and informally. In particular, part of our social interaction will include online gaming sessions. If accepted submissions will not require a full three hours, we are prepared to invite a panel of esports and high-performance HCI researchers to discuss industry needs and research opportunities.

EHPHCI will require accepted authors to synchronously attend its virtual workshop. They may either present their work live, or play prerecorded videos of their presentations for live viewing. In either case, they will answer live moderated questions about their presentations. If necessary, we are prepared to host the workshop with the conferencing tools provided by our own institutions; currently we plan to use Zoom and Discord. We will use automated, live captioning if it is available.

Below is an example schedule for a workshop held on Saturday, May 8. We will adapt it to fit our submissions, and CHI's own conferencing constraints. We have organizers and expect participation

from Asia, North America and Europe, and will therefore begin the workshop in the early morning in North America, afternoon in Europe, and evening in Asia. The times below are eastern US times:

- 8:00** Welcome and organizer introductions
- 8:10** Introductions by first quarter of attendees
- 8:20** Keynote speaker, chaired by Josef Spjut and Ben Watson
- 8:40** Questions for and discussion about keynote
- 8:50** *Ten minute break*
- 9:00** Introductions by second quarter of attendees
- 9:10** *Session 1: Low latency systems*, chaired by Sunjun Kim and Joohwan Kim
 - Paper 1
 - Poster 1
 - Paper 2
 - Poster 2
 - Questions and discussion
- 9:50** *Ten minute break*
- 10:00** Introductions by third quarter of attendees
- 10:10** *Session 2: Training and expertise*, chaired by Byungjoo Lee and Raphael Wimmer
 - Paper 3
 - Poster 3
 - Paper 4
 - Poster 4
 - Questions and discussion
- 10:50** *Ten minute break*
- 11:00** Introductions by last quarter of attendees
- 11:10** *Session 3: Applications*, chaired by Jennifer Listman and David Putrino
 - Case study 1
 - Case study 2
 - Case study 3
 - Case study 4
 - Questions and discussion
- 11:50** Announcements and close of formal program
- 12:00** *Ten minute break*
- 12:10** Parallel gameplay and informal discussion
 - Esports gameplay, e.g. Overwatch
 - Informal discussion: topic elicitation, then breakouts
- 12:50** Close of informal program

8 POST-WORKSHOP PLANS

Some EHPHCI organizers already have strong interest from IEEE Computer Graphics and Applications in an esports position paper. We plan to fold some EHPHCI workshop outcomes into this paper.

We would also like to enable authors to expand their workshop contributions presented at the conference into a journal special issue. We hope to arrange such an issue with either the Proceedings of the ACM on Human-Computer Interaction or the Proceedings of the ACM on Computer Graphics and Interactive Techniques. The organizers will reach out to these journals to propose this option if the workshop is accepted.

Finally, we hope to build and maintain the new EHPHCI community gathered by this workshop. To that end, we will invite

workshop participants to register with a workshop email list (e.g. Google Groups, MailChimp). We will also attempt to identify leadership for a future EHPHCI workshop, and to recruit new organizers and new ideas from this year's attendees.

9 CALL FOR PARTICIPATION

Esports and High Performance HCI This one-day workshop seeks to coalesce and establish a rigorous research effort on supporting expertise in esports and other high-performance applications. Esports is unique in this regard, pushing the envelope of human ability and supporting interfaces, while also producing a wealth of behavioral data. However, a wide range of applications could benefit from such an effort, including rehabilitative medicine, work productivity, professional training, robotic surgery, and more. The study of esports and other applications of high-performance HCI can inform the design of user interfaces, input devices, displays, ergonomic furniture, best practices in communication and group composition, and training protocols.

To begin progress toward these goals, we invite applicants to submit 2-4 page position or research papers (not including references), or one-page posters or case studies. Submissions should address the themes below and be sent to ehphci@googlegroups.com no later than February 21st, 2021. For further submission details on these submission types and their formats, please see the workshop website at ehphci.org.

The workshop will accept those submissions that are highest quality, most relevant, and will generate discussion, as reviewed by workshop organisers. At least one author of each accepted submission must attend the workshop, registering for the workshop and at least one day of the conference.

Topics of interest include, but are not limited to:

- low latency systems
- player performance assessment and training
- expertise
- social impacts
- competition, teamwork and communication
- effect of high-end gaming equipment on performance
- novel game interfaces
- impact of AR/VR on performance
- broader applications of esports technology

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